

# Models

# Kinds of research

- Experimental (we introduce an intervention and look at the effects; researcher introduced assignment)
- Observational (we measure/survey our participants without trying to affect them; no researcher-introduced assignment)

Typically we pair some kinds of statistical tests with experimental work and other kinds of tests with observational work.

# Kinds of research

In reality, most statistical tests can be used with most kinds of research. It's not so much the kind of research that matters, but **which statistic helps to answer your question** and **what types of variables do you have?**

- We'll discuss the first point throughout the course

# What is a model?

not mathematical

# What is a model?

- a representation of the world
- a **statistical** model uses math to make predictions about the world

**YOU DEFINE THE MODEL**

# Middle School Math

$$y = mx + b$$

- what is  $y$ ?
- what is  $m$ ?
- what is  $x$ ?
- what is  $b$ ?

# Let's rewrite this

$$y = b_0 + b_1 X$$

- what is  $y$ ?
- what is  $b_0$ ?
- what is  $b_1$ ?
- what is  $X$ ?

**Are models always right?**

# MODELS ARE FLAWED

- How?

How do we compensate?

$$y = b_0 + b_1X + e$$

**What are the goals of  
modeling?**

**What do you need in order to  
develop a model?**

How do we know if a model is good? What makes it good?

# How will we use models?

- This semester, we will mainly focus on classic statistical tests
- Every single one of these is a model
- We will also focus on developing your intuition
- When you face new models, come back to these basics